

SAVANAH FRANCO DE FREITAS

Forest Engineer

Remote Sensing Division & National Institute for Space Research (INPE)

Tropical Ecosystems and Environmental Sciences Laboratory (TREES)

Remote Sensing II Building, Astronautas Avenue, Jardim da Granja, São José dos Campos, São Paulo, Brazil, ZIP Code: 12227-010

engflorsav@gmail.com | savanah.freitas@inpe.br | Phone: +55 (92) 98417-4934

<http://lattes.cnpq.br/3700520433084636> | <https://orcid.org/0000-0001-6226-1669>

RESEARCH EXPERIENCE

- 2025 - Masters Student at National Institute for Space Research, Tropical Ecosystems and Environmental Sciences Laboratory (TREES)
- Investigating the resilience of secondary Amazonian forests subjected to extreme droughts, through the Amazon-SOS: A Safe Operating Space for Amazonian Forests project (Supervised by Luiz Aragão and Stephen Sitch).
- 2023 - 2024 Research student, Geoprocessing and Remote Sensing Laboratory, Federal University of Amazonas
- Aerophotogrammetry and artificial intelligence to quantify trees and palms for forest inventories in Amazon Rainforests.
- 2019 - 2023 Undergrad Research student, Laboratory of Forest Management, National Institute for Amazonian Research (INPA) and NGEE-Tropics (Next-Generation Ecosystem Experiments)
- Seed bank in an area hit by a convective storm in Central Amazonia;
 - Leaf morphoanatomy of naturally regenerating tree species in an area hit by a convective storm in Central Amazonia;
 - Ecophysiological characteristics of natural regeneration in an area hit by a windstorm (Blowdown) in Central Amazonia;
 - Transpiration dynamics of Terra Firme trees in Central Amazonia.

EDUCATION

- 2025 - M.S. National Institute for Space Research (*Remote Sensing*) – In progress
- 2018 - 2024 B.S. Federal University of Amazonas (*Forest Engineer*)

PROFESSIONAL ACTIVITIES & RECOGNITION

Awards and Honors

- 2024 National Institute for Amazonian Research (INPA), Brazil, 2022–2023, Project: Seed bank in an area affected by a convective storm in the Central Amazon, Congress of Institutional Scientific Initiation Scholarship Program.
- 2024 Climate and Forests Study Group (GECLIF), Federal University of Amazonas (UFAM) and Federal Institute of Amazonas (IFAM), 2023–2024, Project: PodFlor - Accessible Forests and Environment for Visually Impaired People, National Congress on Amazon Research.

Professional/Student Involvement

- 2025 - Independent consultant at CTrees
- Sampling for Land Use and Land Cover (LULC) and environmental degradation.
- 2025 - 2026 Independent consultant at GFZ Helmholtz-Zentrum für Geoforschung (Nov 2025 – Apr 2026)
- Conducted spatial sampling of forest cover and degradation areas for dataset generation.
- 2024 - 2024 Research Collaborator, Laboratory of Forest Management, National Institute for Amazonian Research (INPA)
- Technician and research supporter at Madeiras da Amazônia, a National Program of Science and Technology;
 - Instructor at the Biomass Estimation Course (Water and Carbon) in the Amazon Rainforest, focusing on Sustainable Forest Management and Climate Change;
 - Technician and research supporter at NGEE Tropics.
- 2023 - Climate and Forests Study Group (GECLIF)
- PodFlor - Accessible Forests and Environment for Visually Impaired People;
 - Scientific dissemination and forest sciences: the experience of the Kaapeguara Project;
 - Wood technology in the training of young luthiers in the city of Manaus: the Jurupari Project;
 - Web-Xiloteca: Implementation of a Digital Repository for an Amazonian wood collection;
 - Florestando: A digital repository about the Amazon for children.
- 2022 - 2024 Center for Geotechnologies applied to the Environment (NUGEA), Federal University of Amazonas (UFAM)

- ARP images to estimate Biomass in the Amazon using algorithms and open data;
- Study of the potential for quantifying carbon stock estimates in forests using low-cost aerophotogrammetry;
- Aerophotogrammetry and artificial intelligence to quantify trees and palms in Amazon native Rainforests;
- Remote Sensing and GIS Analyst for research projects.

2021 - 2023 Forest Exploration Laboratory, Federal University of Amazonas (UFAM)

- Instructor to undergraduate students at remote sensing for forest exploration in tropical regions.

2021 - 2023 EMCOF Junior Enterprise

- Remote sensing and GIS instructor and analyst for forest management.

Languages

PORTUGUESE.	Native tongue.
ENGLISH.	Good reading, writing, speaking and listening skills.
GERMAN.	Basic level.
SPANISH.	Basic level.